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MINISTRY OF MINES

# Mining Sector Medium-Term (2019-2021 E.C.) Development and Investment Plan

## የማዕድን ዘርፍ የመካከለኛ ዘመን (2019-2021) የልማትና የኢንቨስትመንት ዕቅድ

**Megabit 2018 E.C. (March/April 2026 G.C.)**

Addis Ababa, Federal Democratic Republic of Ethiopia

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# Part One: Introduction and Strategic Orientation

## 1. Introduction

The Ministry of Mines is mandated to formulate and implement policy and legal frameworks governing the sector, conduct geoscience investigations to assess mineral deposits so that the mining sub-economy contributes its rightful share to national economic growth, license and support investors entering the sector, ensure the extensive and qualitative study and development of the sector, produce foreign-exchange-generating minerals, and substitute imported mineral products and industrial mineral inputs in the domestic market to play an indispensable role in generating and conserving foreign exchange.

Furthermore, the Ministry is tasked with ensuring that sector development is conducted in a manner that protects environmental and community safety, creating conducive conditions for local residents in mining corridors to benefit from employment opportunities and public infrastructure, and collaborating with relevant stakeholders to organize and build the capacity of artisanal mineral producers to serve as a sustainable source of national capital. To realize these mandates, the Ministry has entered into operation guided by the strategic goals and roadmaps set out in the national 10-Year Development Plan.

Over the past years, the comprehensive macroeconomic reform implemented across Ethiopia has made a significant positive contribution and unlocked extraordinary opportunities for the mining sector. By curbing illegal trading and contraband smuggling that previously drained gold produced by artisanal and special small-scale miners, and directing improved volumes through legal banking channels to the National Bank of Ethiopia (NBE), unprecedented foreign exchange receipts have been secured from the sector.

Concurrently, to utilize the country's natural resource endowments at an elevated scale, an enabling investment climate has been established, attracting developers with substantial capital, advanced technology, and high operational capability into gold, iron ore, potash, and associated minerals.

To operationalize mega mining enterprises that hold substantial potential for foreign exchange generation and import savings, multiple legal, regulatory, and procedural improvements were instituted. Through rigorous follow-up and institutional problem-solving, mega companies such as Ethno, YMG, Pioneer, Lemi National Cement, and ET have successfully entered commercial production. Looking ahead, additional high-capacity enterprises in gold, ceramics, and granite are finalizing pre-production works and will commence full operations in the short term.

In the petroleum and gas sub-sector, the first phase of the Ogaden Liquid Natural Gas (LNG) project at Calub in the Somali Region—with a designed annual production capacity of 111 million liters—was officially inaugurated and commenced operations. Concurrently, construction of the second-phase plant with an annual capacity of 1.33 billion liters is actively advancing. In addition, construction has commenced on a 1,000 MW gas-fired thermal power generation project fueled by natural gas output.

Furthermore, preliminary preparatory works are progressing to launch the Gode Oil Refinery project, designed to extract and refine crude oil from the Hilala field with an annual targeted capacity of 3.5 million tons. In parallel, activities are underway for the Dangote Gode Fertilizer Factory project—undertaken jointly by Ethiopian Investment Holdings and Dangote Group—to utilize Calub natural gas to produce and supply 3 million tons of urea annually to the domestic and regional market.

In the upcoming final phase of the investment plan, recognizing that past fragmented studies left gaps in understanding Ethiopia's mineral potential via modern technological capabilities, priority is placed on deploying airborne geophysical surveys across broad territories to drastically compress multi-year exploration timelines and obtain multi-purpose geospatial intelligence. Additional core priorities include institutional capacity building through service digitalization; scaling the variety, quality, and volume of export minerals to boost foreign currency receipts; expanding domestic substitution of industrial mineral inputs and outputs; conducting modern technology-backed geological mapping; enhancing project contract administration and compliance; transitioning pre-production concessionaires into commercial output; generating citizen employment; ensuring environmental sustainability and local community benefit-sharing; and collaborating seamlessly with critical national stakeholders.

This document reviews the achievements and challenges encountered during the first three years (2016-2018 E.C.) of the 10-Year Development Plan, and sets forth the medium-term (2019-2021 E.C.) development implications, expected results, strategic goals, core activities, key targets, and medium-term capital investment requirements, prepared in accordance with the national "One Plan, One Report" preparation and presentation guidelines formulated by the Ministry of Planning and Development.

## Vision of the Mining Sector

**To see our mineral wealth serve as the foundation of national prosperity by 2022 E.C.**
*(በ2022 የማዕድን ሀብታችን የብልፅግና መሠረት ሆኖ ማየት)*

## Mission of the Mining Sector

**To develop our mineral resources and transform them into a sustainable source of national capital.**
*(የማዕድን ሀብታችንን በማልማት ለሀገራዊ ካፒታል ምንጭ ማድረግ)*

## Core Values of the Mining Sector

| | | | |
|---|---|---|---|
| **Diligence (ትጋት)**<br>Unwavering commitment to excellence | **Collaboration (ትብብር)**<br>Synergistic institutional partnerships | **Integrity / Transparency (ንፅህና)**<br>Ethical stewardship & public trust | **Service Excellence (አገልግሎት)**<br>Dedicated citizen & developer support |

# Part Two: Three-Year Development Plan Performance (2016-2018 E.C.)

## 2.1 Ensuring the Country's Geological Information Coverage, Quality, and Accessibility

Over the past investment years, fundamental geoscience, geochemical, geophysical, and geological hazards studies were carried out to improve the coverage, quality, and accessibility of national geological data. Exploration focused on identifying minerals that boost agricultural productivity, supplying industrial feedstocks, gathering baseline infrastructural data, and prospecting for foreign-exchange-generating and import-substituting minerals. Across 12 target areas, 1:50,000 scale geological and geochemical mapping covered 17,250 sq. km with complete thematic maps and technical reports. Furthermore, 6 prospective geothermal sites covering 6,000 sq. km were delineated for direct heat utilization.

Resource identification and reporting were finalized for placer gold, chromite/cobalt, lithium/tantalum, industrial minerals, nickel, gemstones, and coal; 42 thoroughly studied mineral prospective sites were officially transferred to regional administrations for development.

The historical 1:2,000,000 scale geological map of Ethiopia, which served for decades, lacked recent findings derived from modern technologies. To modernize the national geological map at a more detailed resolution, standardization and harmonization initiatives were executed in partnership with Addis Ababa University.

Previously, extensive geoscience data generated by the Geological Institute of Ethiopia and high-value drilled core samples were stored without adequate archival standards, impeding client access. To resolve this bottleneck, the National Geoscience Data Center Initiative was launched. The National Geoscience Data Center and Mini Gallery construction was completed, organizing decades of historical geological and mineral exploration reports and maps to professional standards. Concurrently, data conversion from raster images to digital GIS formats is underway, alongside construction of the dedicated Core Archive repository building.

The existing geoscience laboratory had operated for over 30 years with aging infrastructure and equipment unsuited to modern requirements. To modernize testing services and establish an internationally accredited laboratory capable of testing complex samples domestically—thereby curbing foreign exchange outflows—a rehabilitation contract was signed with the Ethiopian Engineering Corporation, with reconstruction now 67% complete.

To support geological research and subsurface exploration, new drilling equipment was procured to replace aging rigs that had fallen out of service, directly reinforcing ongoing mineral reserve estimation projects. Across the review period, a total of 113,051 geochemical and mineralogical sample analyses were performed, and 39,356 geoscience data sets were disseminated to stakeholders and clients.

## 2.2 Expanding and Growing Domestic and Foreign Investment in Mining and Petroleum

Over the past three years (2016-2018 E.C.), continuous Mining and Technology Expos were organized to promote Ethiopia's mineral potential, drawing participation from mining companies, gemstone producers/exporters, equipment manufacturers/importers, and financial institutions. The expos facilitated direct business-to-business links between local producers and international investors, supplemented by panel discussions highlighting investment opportunities.

Across the plan period, 81 mining licenses were issued—63 to domestic investors and 18 to foreign investors—spanning exploration, extraction, and processing, mobilizing an aggregate registered investment capital of 653.472 billion Birr. Additionally, 4 geothermal well drilling permits were granted, and licenses were issued to 25,109 artisanal mining enterprises.

Through concerted engagement between the National Mining Council and key stakeholders, comprehensive regulatory and operational adjustments were introduced. Favorable operating environments enabled mega enterprises with substantial foreign exchange generation, saving, and import substitution potential—including Ethno, YMG, Pioneer, Lemi National Cement, and ET—to enter commercial production.

In the Somali Region (Calub), Phase 1 of the Ogaden Liquid Natural Gas (LNG) project (111 million liters/year capacity) was commissioned, and key downstream infrastructure was established to receive output. Construction of the Phase 2 plant (1.33 billion liters/year) is advancing with ongoing equipment manufacturing and on-site civil works at Calub. Construction also commenced on the 1,000 MW gas-fired thermal power project.

Preparatory milestones were advanced for the Gode Oil Refinery (3.5 million tons/year refining capacity from Hilala crude). Furthermore, progress continued on the Dangote Gode Fertilizer Factory project (undertaken by Ethiopian Investment Holdings and Dangote Group) to manufacture 3 million tons of urea annually using Calub natural gas.

### Import Substitution Performance

To strengthen the construction sector, reduce structural dependence on imported building supplies, and enhance the sector's macroeconomic contribution, targeted interventions were prioritized under the 10-Year Plan. Structured coordination platforms were instituted with key stakeholders to monitor and support capacity utilization, product quality, and productivity across cement, coal, steel, ceramics, marble/granite, and glass industries.

#### Coal Development and Substitution

To substitute imported coal heavily utilized by cement plants as an energy source at high foreign exchange expense, the Ministry intensified technical support and oversight across licensed coal processors, successfully operationalizing processing plants including ET, Yo Holding 2, and National Mining. Field assessments were conducted on operational coal producers and coal-washing plants, culminating in an institutional partnership with Addis Ababa University to address operational challenges. Deliverables finalized under this partnership include the Ethiopian Coal Strategic Document, a 10-Year Coal Roadmap, mining and design plans, quality control protocols, value chain logistics assessments, environmental/health/safety standards, governance frameworks, training manuals, operational guidelines, quality standards, and a centralized coal database management system.

### Cement Industry Performance

To meet expanding construction demand, strong monitoring and facilitation enabled the commissioning of new facilities, prominently the Lemi National Cement Factory (production capacity of 150,000 quintals per day) and the expansion of Pioneer Cement (raising capacity from 400,000 to 600,000 tons/year). This significantly improved market supply and stabilized domestic cement prices.

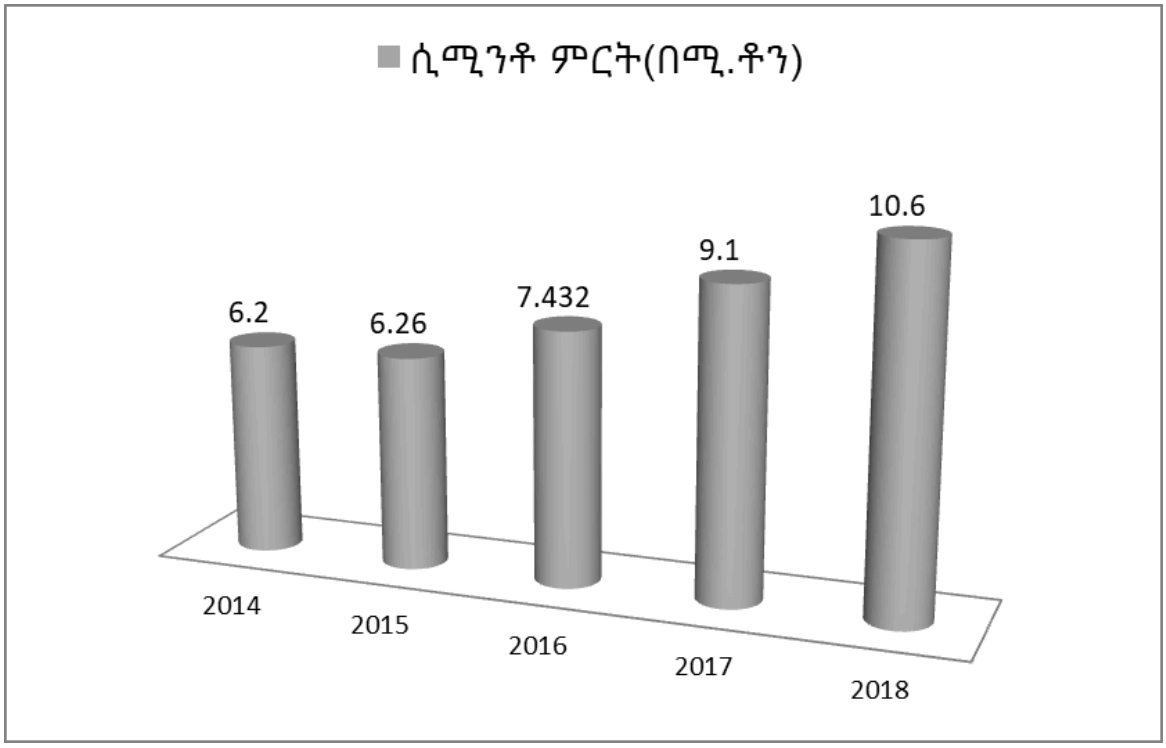



*Figure 1: National Cement Production Growth (in Million Quintals: 2014: 6.2M, 2015: 6.26M, 2016: 7.432M, 2017: 9.1M, 2018: 10.6M quintals).*

### Iron and Steel Industry

Recognizing steel imports as a major foreign exchange drain, high priority was placed on domestic substitution, starting with reserve identification. Sector assessments identified critical domestic bottlenecks: local raw material shortages, fragmented inter-agency coordination, shortage of skilled human capital and technology transfer, and non-standardized finished goods. To provide temporary relief for billet shortages, institutional scrap metal inventories were surveyed and allocated to rolling mills. For long-term sustainable supply, iron ore exploration and reserve estimation partnerships were established with Addis Ababa University, conducting detailed geological exploration across 1,000 sq. km in the Sekota area of Northern Ethiopia, while facilitating the entry of qualified developers into the steel value chain.

### Ceramics and Dimensional Stone

To support construction, multiple factories currently produce floor/wall ceramic tiles and sanitary ware. The Badjea Ceramic factory is projected to produce 35,000 sq. meters per day during its initial production run, utilizing 95% local raw material inputs. The Grandeur Ceramic project is being established with a designed capacity of 16,000 sq. meters per day to substitute large-format imported tiles. Furthermore, the CAM Ceramic Products manufacturing project is designed to produce 18,400 sq. meters per day upon completion. In addition, the Marble Center of Excellence and capacity-building hub is under active construction at Assosa University.

#### Agricultural Inputs Development

Soil acidity mapping and limestone resource clustering were conducted across affected agricultural belts to support agricultural lime production. A 10-Year Strategic Document and operational training manual for agricultural lime development were prepared.

### Mining Sector Domestic Revenue and Employment Generation

The formal mining sector generated direct employment for 4,938 citizens, while regional artisanal mining enterprises engaged 663,604 individuals (415,543 male and 248,061 female), resulting in an aggregate 668,542 jobs created over the three-year plan period.

Domestic revenue mobilization reached 1,930.5 million Birr from mining royalties, 5.241 million Birr from miscellaneous federal regulatory fees (licensing, registration, renewals, and competency certifications), and 9.334 billion Birr collected across regional states from diverse mining operations.

## 2.3 Strengthening Local Community Linkages in Mining and Petroleum Investment

Ensuring equitable community benefits is vital for sustainable sector development. The Ministry enforced contractual compliance, collecting 115.5 million Birr from mining and petroleum concessionaires for local community development funds. Of this, 102.65 million Birr was disbursed to regional states, financing health posts, potable water tanker trucks, hospital medical supplies, school and farmers' training center construction, and feeder road rehabilitation. In parallel, environmental rehabilitation was completed across 109 hectares of post-mining land.

## 2.4 Export Minerals Production and Foreign Exchange Earnings

### Gold Production Performance

The comprehensive macroeconomic reform introduced at the start of the 2017 fiscal year exerted a transformative impact on the gold sector. Previously, extensive gold mined by artisanal and special small-scale miners was diverted through illicit trade and contraband networks. Policy adjustments and competitive purchasing incentives introduced by the National Bank of Ethiopia channeled massive volumes into the formal financial system, yielding historic foreign currency receipts.

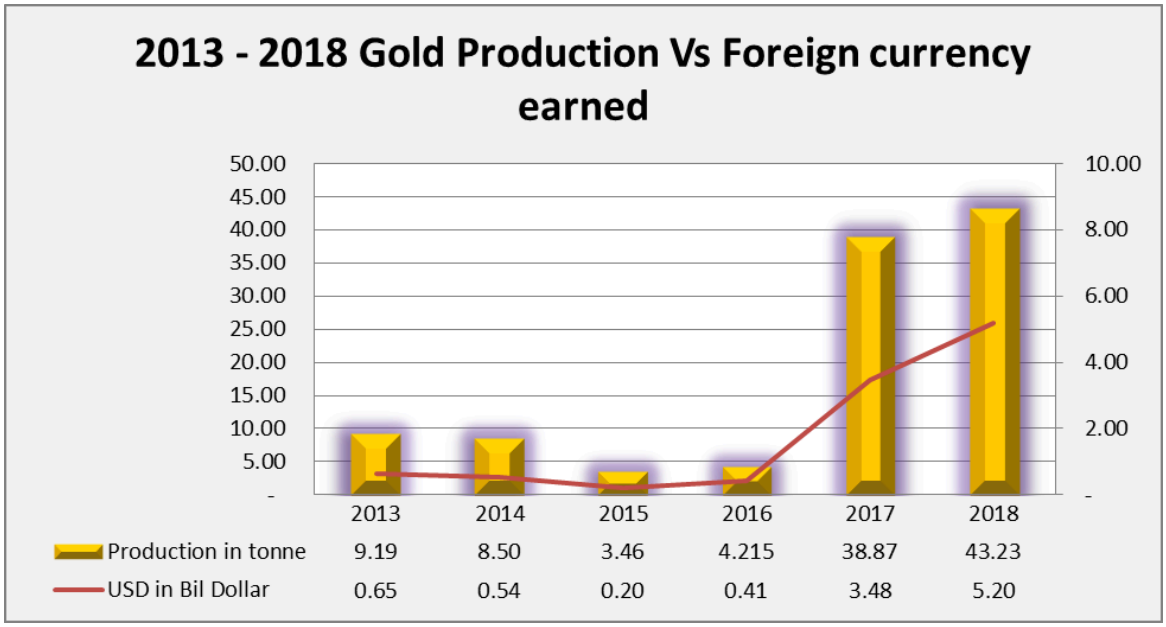



*Figure 2: 2013 - 2018 Gold Production (tonnes) vs. Foreign Currency Earned (Billion USD).*

While gold export earnings and supply had sharply declined between 2014 and the end of 2016 due to contraband smuggling and security constraints, formal receipts escalated dramatically following the 2017 reform. Total foreign exchange generated from gold exports reached USD 3.48 billion in FY 2017. In FY 2018, receipts reached USD 3.51 billion within the initial nine months and are projected to attain USD 5.20 billion by the close of the fiscal year. Over the cumulative period of the 10-Year Plan to date, 107.46 tons of gold were delivered to the National Bank of Ethiopia against a planned target of 63.33 tons—representing an outstanding 170% plan achievement.

### Corporate Gold Production (MIDROC Gold)

MIDROC Gold maintained consistent commercial production over consecutive years. To sustain and expand output, joint oversight and support mechanisms are maintained. Looking forward, the planned entry of high-capacity commercial developers—such as Kurmuk Gold Mine—alongside mid-tier concessionaires is projected to structurally transform Ethiopia's gold sector from overwhelming dependence on artisanal output to an industrialized production base.

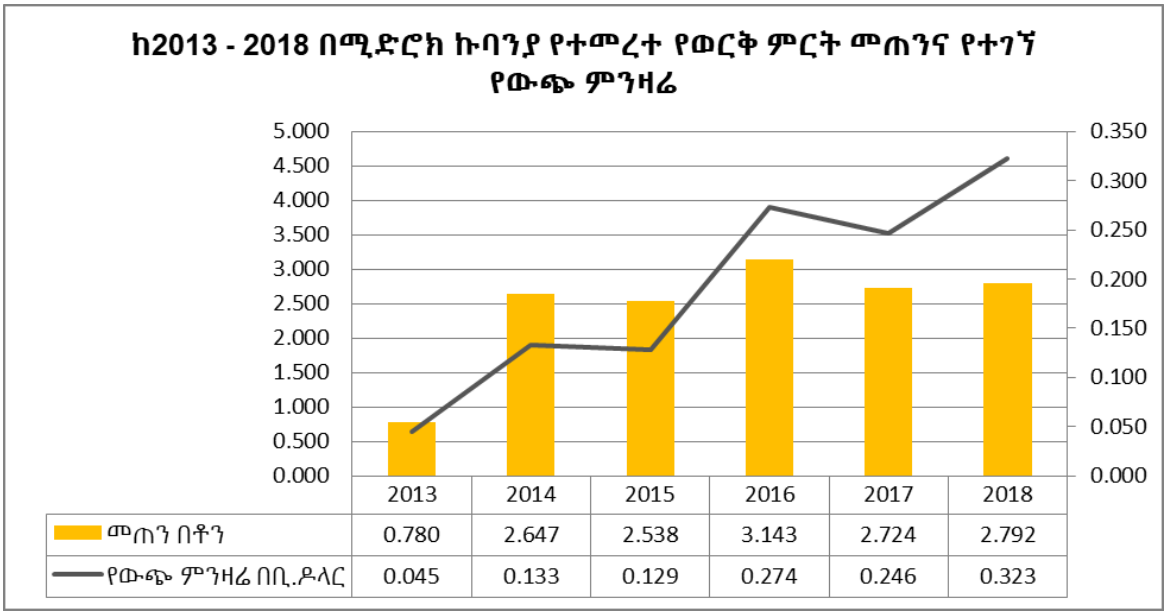



*Figure 3: 2013 - 2018 MIDROC Gold Production Volume (tonnes) and Foreign Currency Receipts (Billion USD).*

### Artisanal Gold Production Performance

From 2013 to 2015, formal artisanal gold deliveries to the central bank remained depressed. Following the 2017 macroeconomic reforms, artisanal and small-scale miners experienced unprecedented mobilization, expanding deliveries by more than fourfold. To sustain this momentum into the 2019-2021 investment cycle, collaborative programs with all regional states will expand capacity building, modern processing tools, occupational safety, and technical training. Concurrently, rigorous monitoring will ensure licensed small-scale corporate miners meet their contractual performance thresholds.

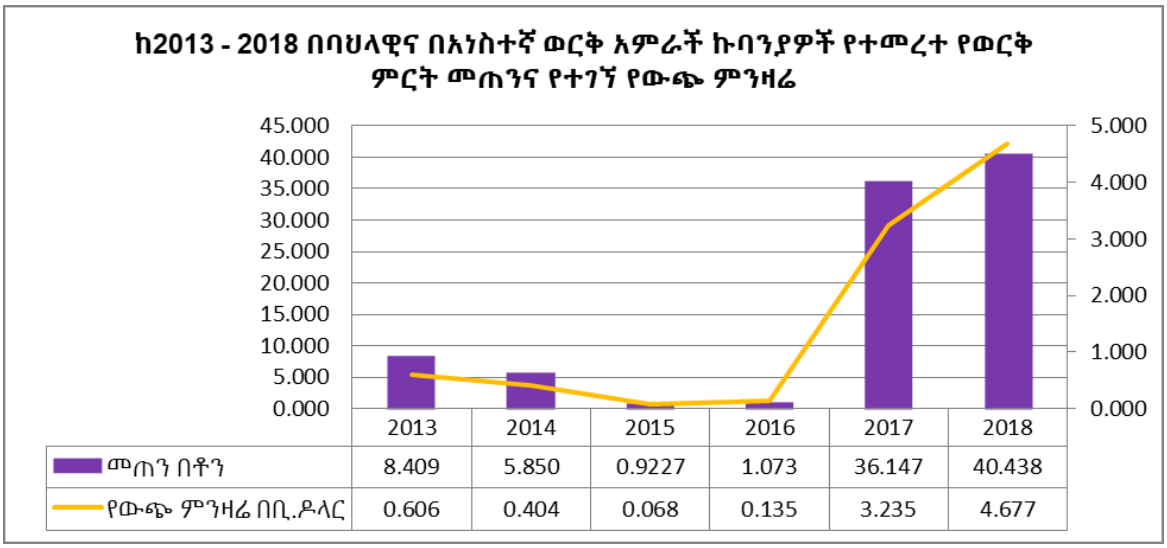



*Figure 4: 2013 - 2018 Artisanal Gold Production Volume (tonnes) and Foreign Exchange Earned (Billion USD).*

## Opal Production Performance

Opal is supplied to the export market in both rough (raw) and value-added processed forms.

### Raw Opal Performance and Price Evolution

Export volume shifted from 22,031.22 kg in FY 2016 to 8,738.03 kg in FY 2017 (a 60% volume contraction). However, gross revenue increased from USD 2.00 million to USD 2.267 million as international market

prices appreciated. The average unit price escalated from USD 90.78/kg in 2016 to USD 259.49/kg in 2017 (a 186% year-on-year increase), reaching USD 279.76/kg in the first nine months of FY 2018. This demonstrates a strategic market shift from high-volume/low-value exports toward premium-priced supply.

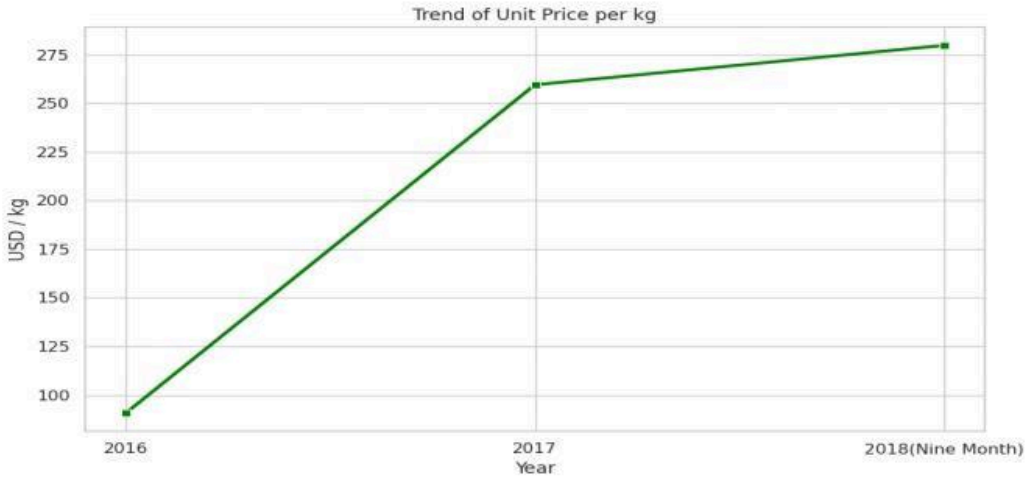



*Figure 5: Trend of Raw Opal Export Unit Price per Kilogram (USD/kg, 2016-2018).*

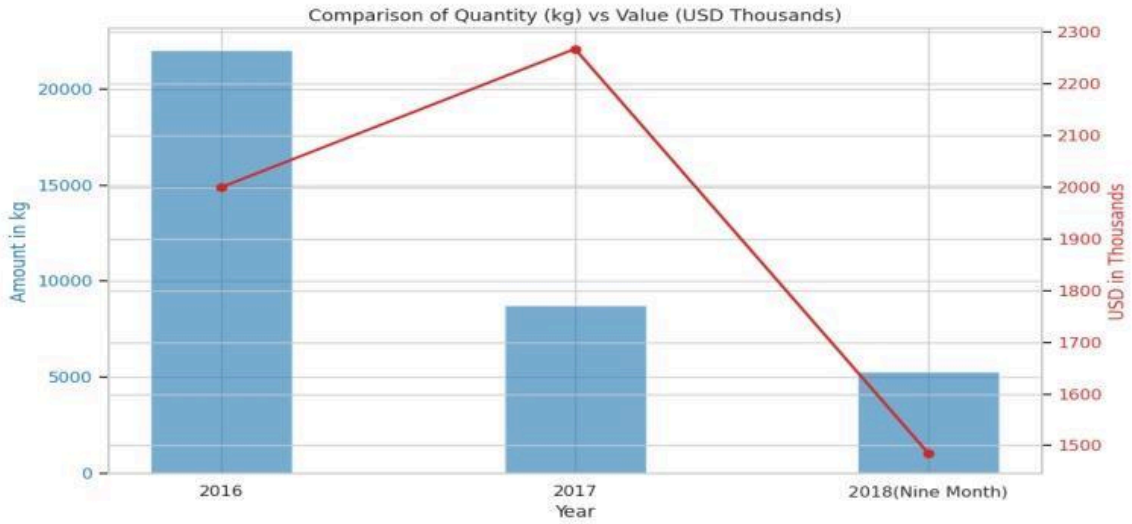



*Figure 6: Comparison of Raw Opal Export Quantity (kg) vs. Value (USD Thousands).*

### Value-Added Opal and Gemstone Commercialization

Export earnings expanded from USD 751,000 in 2016 to USD 1,133,700 in 2017 (an approximate 50% revenue growth) despite a 13% decline in shipment volume, as unit prices increased from USD 1,633.50/kg to USD 2,821.13/kg.

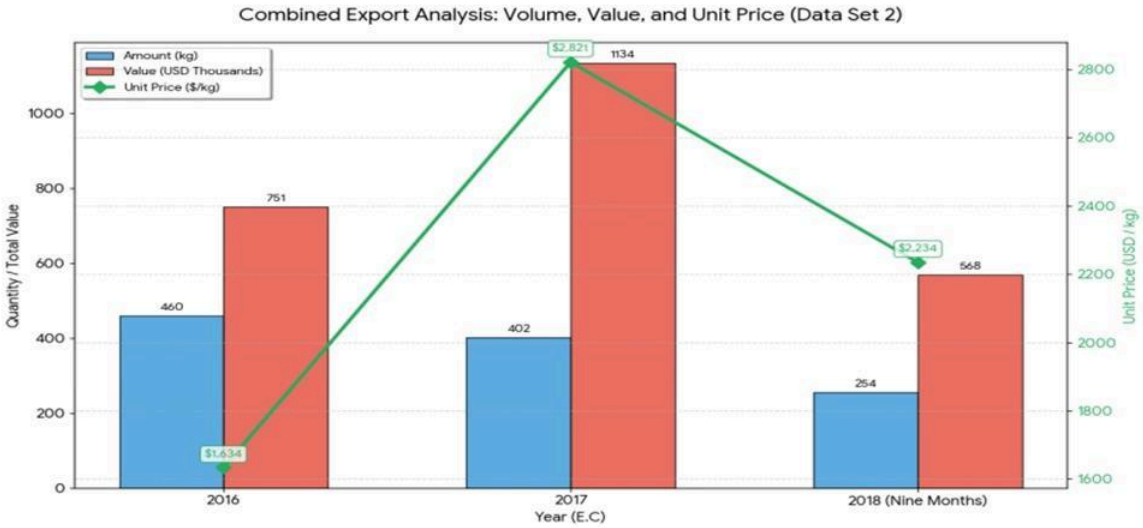



*Figure 7: Combined Export Analysis: Volume, Value, and Unit Price for Value-Added Opal.*

Web portals, websites, and mobile applications are under active development to regulate, promote, identify, and grade Ethiopian gemstones. Concurrently, integration of opal trading into the Ethiopian Commodity Exchange (ECX) modern trading system is near finalization, supported by draft trading directives. In partnership with Wollo University, specialized technological prototypes—including soil-moving carts, gemstone cutting and polishing machines, and waste rock transport equipment—are finalizing engineering fabrication.

## Industrial Minerals Export Performance

Despite volume contractions in FY 2017, surging unit values sustained growth in aggregate export earnings. The unit price of industrial minerals increased to USD 388.50 per ton in 2018, reflecting a 45% increase over 2016 levels. Total earnings generated during the first nine months of FY 2018 surpassed the full-year totals of both prior years combined.

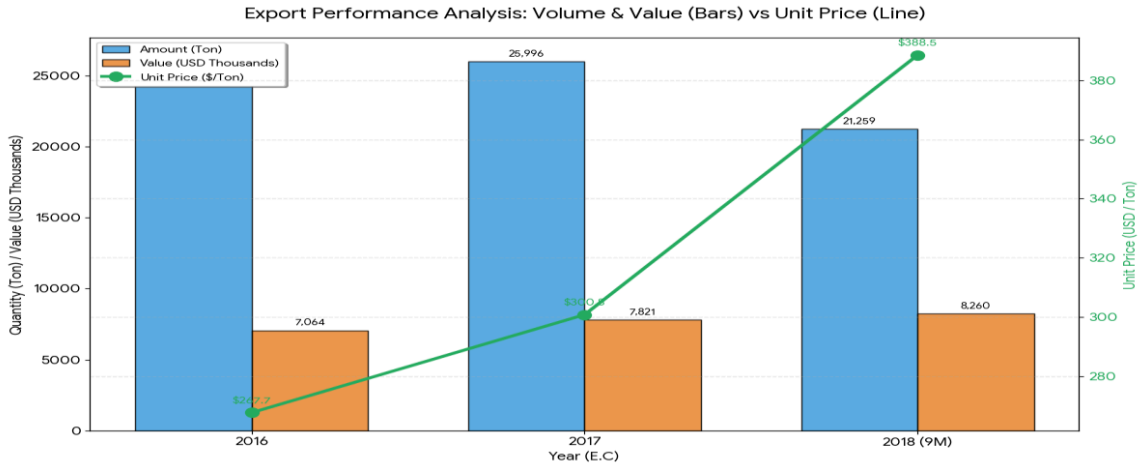



*Figure 8: Export Performance Analysis: Volume & Value (Bars) vs Unit Price (Line) for Industrial Minerals.*

## 2.5 Mining Sector (2016-2018) Three-Year Development Plan Performance Table

| No. | Strategic Goals, Main Activities, and Key Performance Indicators | Unit of Measurement | 2016 | 2017 | 2018 |
|---|---|---|---|---|---|
| **1** | **Ensuring the Country's Geological Information Coverage, Quality, and Accessibility** | | | | |
| ***1.1*** | ***Enhancing Geoscience, Geochemistry, Geophysics, and Geohazards Coverage, Quality, and Accessibility*** | | | | |
| 1.1.1 | Fundamental geoscience mapping at 1:50,000 scale (Number of map sheets) | Number | 4 | 4 | 4 |
| 1.1.2 | Fundamental geoscience mapping coverage at 1:50,000 scale | Sq. Km | 5,250 | 600 | 6,000 |
| 1.1.3 | Identification of geothermal prospective sites for direct utilization | Number | 2 | 2 | 2 |
| 1.1.4 | Exploration coverage of geothermal sites for direct utilization | Sq. Km | 3,000 | 3,000 | 3,000 |
| ***1.2*** | ***Exploration and Study of Mineral Inputs Enhancing Agricultural Productivity*** | | | | |
| 1.2.1 | Identification of mineral reserve sites suitable for fertilizer inputs | Number | - | 1 | 1 |
| 1.2.2 | Exploration coverage for fertilizer mineral inputs | Sq. Km | - | 750 | 750 |
| 1.2.4 | Production of weed and pest control agrochemicals | Liters | 475,559 | 348,341 | 142,942.5 |
| 1.2.5 | Supply of agricultural limestone to mitigate soil acidity | Million Quintals | 0.28 | 0.64 | 0.702 |
| ***1.3*** | ***Expanding Exploration and Studies for Infrastructure and Construction Industry*** | | | | |
| 1.3.1 | Iron ore prospective exploration and reserve estimation studies | Reserve Studies (No.) | - | 1 | 1 |
| 1.3.2 | Nickel mineral exploration and reserve estimation studies | Reserve Studies (No.) | - | - | - |
| 1.3.3 | Coal mineral reserve estimation studies | Reserve Studies (No.) | - | - | - |
| 1.4.5 | Iron ore prospective exploration and study | Site Count | - | 1 | - |
| 1.4.6 | Coal prospective exploration and study | Site Count | 1 | 3 | 2 |
| 1.3.4 | Chromium / Cobalt prospective exploration and study | Site Count | 2 | 2 | 2 |
| 1.3.5 | Nickel prospective exploration and study | Site Count | 2 | 2 | 2 |
| ***1.4*** | ***Expanding Exploration and Studies of Export-Oriented Minerals*** | | | | |
| 1.4.1 | Placer gold prospective exploration and study | Site Count | 2 | - | - |
| 1.4.2 | Gemstone prospective site exploration and study | Site Count | 2 | 2 | 2 |

| No. | Strategic Goals, Main Activities, and Key Performance Indicators | Unit of Measurement | 2016 | 2017 | 2018 |
|---|---|---|---|---|---|
| 1.4.3 | Tantalum and Lithium prospective site exploration and study | Site Count | 2 | 3 | 2 |
| 1.4.4 | Industrial minerals prospective site exploration and study | Site Count | 3 | 3 | 3 |
| ***1.5*** | ***Expanding Geological Survey Extension Services to Regional States*** | | | | |
| 1.5.1 | Transfer of delineated mineral potential sites to regional states | Number | 17 | 16 | 11 |
| 1.5.2 | Capacity-building mineral exploration training delivered to regions | Number | - | 4 | 4 |
| 1.5.3 | Geoscience research deliverables disseminated to users | Data Sets (No.) | 12,856 | 13,500 | 13,000 |
| 1.5.4 | Geoscience laboratory samples analyzed and tested | Number | 27,636 | 55,350 | 30,065 |
| **2** | **Enhancing Private Sector Investment and Mining Industry Production Capacity** | | | | |
| ***2.1*** | ***Expanding Domestic and Foreign Investment in Mining and Petroleum*** | | | | |
| 2.1.1 | Exploration, production, and processing licenses issued to domestic investors | Number | 13 | 35 | 30 |
| 2.1.2 | Exploration, production, and processing licenses issued to foreign investors | Number | 8 | 8 | 10 |
| 2.1.3 | Capital registered from newly issued mining and geothermal licenses | Million Birr | 4,955 | 4,911 | 653.472 |
| 2.1.4 | Number of artisanal mineral producer enterprises organized | Number | 8,385 | 10,698 | 9,000 |
| ***2.2*** | ***Problem-Solving Industrial Studies Enhancing Mining Production Capacity*** | | | | |
| 2.2.1 | Feasibility study conducted for fertilizer resource development | Number | - | 1 | - |
| 2.2.2 | Feasibility study documents prepared for industrial mineral inputs | Number | - | - | - |
| 2.2.3 | Supply chain and distribution bottleneck studies conducted for coal | Number | - | - | - |
| 2.2.4 | Baseline production assessments of artisanal mining operations | Number | 2 | 3 | 5 |
| 2.2.4b | Adapted, modified, and scaled mineral technologies | Number | 3 | - | 2 |
| 2.2.5 | Capacity-building training sessions delivered to enhance industrial productivity | Number | 9 | 15 | 20 |

| No. | Strategic Goals, Main Activities, and Key Performance Indicators | Unit of Measurement | 2016 | 2017 | 2018 |
|---|---|---|---|---|---|
| 2.2.6 | Development of natural-gas-based fertilizer projects | Number | - | - | 1 |
| 2.2.7 | Gold production expansion rate | Percentage (%) | - | 8% | - |
| 2.2.8 | Cement industrial capacity utilization rate | Percentage (%) | 55% | 61.5% | 67% |
| 2.2.9 | Steel industrial capacity utilization rate | Percentage (%) | 10.6% | 11.45% | 15.44% |
| 2.2.10 | Ceramic industrial capacity utilization rate | Percentage (%) | 56% | 60.4% | 65% |
| 2.2.11 | Coal processing industrial capacity utilization rate | Percentage (%) | - | - | 70% |
| 2.2.12 | Chemical industrial capacity utilization rate | Percentage (%) | 45% | 48% | 55% |
| 2.2.13 | Glass industrial capacity utilization rate | Percentage (%) | 50% | 52% | 60% |
| ***2.3*** | ***Expanding Mining Industrial Production*** | | | | |
| 2.3.1 | Industrial cement production volume | Million Tons | 9.10 | 6.228 | 10.60 |
| 2.3.2 | Industrial steel production volume | Million Tons | 0.50 | 0.330 | 0.504 |
| 2.3.3 | Industrial ceramic tile production volume | Million Sq. M | 9.47 | 9.12 | 13.97 |
| 2.3.4 | Sheet glass production volume | Sq. Meters | 678,586 | 730,135 | 1,251,000 |
| 2.3.5 | Industrial marble production volume | Thousand Sq. M | 83.0 | 58.19 | 87.70 |
| 2.3.6 | Domestic coal production volume | Million Tons | 2.10 | 1.15 | 1.97 |
| 2.3.7 | Industrial chemical products volume | Metric Tons | 29,763 | 3,340.74 | 5,010 |
| 2.3.8 | Construction mineral raw materials supply volume | Million Cu. M | 72.54 | 34.857 | 36.60 |
| 2.3.9 | Industrial mineral inputs supply volume | Million Tons | 15.35 | 8.85 | 9.29 |
| **3** | **Expanding Mining Sector Output, Foreign Exchange, and Employment** | | | | |
| ***3.1*** | ***Increasing Export Mineral Production*** | | | | |
| 3.1.1 | Gold export production volume | Metric Tons | 4.215 | 38.87 | 43.23 |
| 3.1.2 | Tantalum export production volume | Metric Tons | 131.61 | 174.50 | 122.82 |
| 3.1.3 | Value-added and raw opal export production volume | Metric Tons | 22.322 | 9.780 | 8.886 |
| 3.1.4 | Gemstone export production volume | Metric Tons | 81.90 | 24.577 | 18.00 |
| 3.1.5 | Export-quality marble production volume | Cubic Meters | - | - | 472.69 |
| 3.1.6 | Industrial minerals export shipment volume | Metric Tons | 36,103.44 | 25,995.80 | 27,573.75 |
| 3.1.7 | Foreign exchange revenue generated from mineral export sales | Million USD | 427.42 | 3,450.00 | 5,200.00 |

| No. | Strategic Goals, Main Activities, and Key Performance Indicators | Unit of Measurement | 2016 | 2017 | 2018 |
|---|---|---|---|---|---|
| *3.2* | ***Domestic Import Substitution and Industrial Feedstock Supply Growth*** | | | | |
| 3.2.1 | Number of imported mineral commodities substituted by domestic supply | Count | - | 1 | 1 |
| 3.2.2 | Total volume of substituted mineral products | Million Tons | 0.574 | 2.10 | 1.97 |
| *3.3* | ***Mobilizing Domestic Sector Revenue*** | | | | |
| 3.3.1 | Royalty revenue collected from mineral production | Million Birr | 250.70 | 627.80 | 1,052.00 |
| 3.3.2 | Revenue collected from licensing, registration, renewal, and competency fees | Million Birr | 1.094 | 2.137 | 2.010 |
| 3.3.3 | Miscellaneous mining revenues collected by regional states | Million Birr | 2,382.08 | 3,751.99 | 3,200.00 |
| 3.3.4 | Income tax revenue generated from the mining sub-economy | Million Birr | 95.57 | 1,300.00 | 100.00 |
| *3.4* | ***Job Creation in the Mining Sector*** | | | | |
| 3.4.1 | Jobs created within formal mining companies and processing industries | Number | 826 | 2,112 | 2,000 |
| 3.4.2 | Jobs created within regional artisanal mining enterprises | Number | 248,851 | 243,163 | 171,590 |
| **4** | **Strengthening Community Linkages in Mining and Petroleum Investment** | | | | |
| *4.1* | ***Mobilizing Community Development Funds from Developers*** | | | | |
| 4.1.1 | Community development funds mobilized from mining and petroleum firms | Million Birr | 34.50 | 38.00 | 43.00 |
| 4.1.2 | Percentage of licensed firms fulfilling community development agreements | Percentage (%) | 68% | 89% | 93% |
| 4.1.3 | Community development funds disbursed to regional administrations | Million Birr | 27.38 | 33.27 | 42.00 |
| 4.1.4 | Land area rehabilitated following mining operations | Hectares | 19 | 45 | 45 |

## 2.6 Strengths, Lessons Learned, Challenges Encountered, and Solutions Taken

### 2.6.1 Strengths Identified and Lessons Learned

- **Historic Export Performance**: Remarkable, highly encouraging foreign exchange earnings secured from mineral exports, led by gold.
- **Governance Frameworks**: Core strategic policy roadmaps and operational directives formulated to guide sector administration.

- **Domestic Import Substitution**: Meaningful operational milestones achieved in replacing imported coal with domestic processed coal.
- **Capital Capacity Projects**: Substantive modernization initiatives launched across laboratories, archives, and regional bureaus.
- **Executive Problem-Solving for Mega Concessionaires**: Coordinated federal-regional task forces resolved developer bottlenecks, enabling landmark investments (Ethno, YMG, Pioneer, Lemi National Cement, ET) to enter commercial production.
- **Commercial Readiness in Petroleum & Gas**: Natural gas development progressed to supply feedstocks for national fertilizer and power projects.
- **Large-Scale Concession Inflows**: Substantial international private capital attracted into gold, iron ore, and potash.
- **Raw Material Security for Steel**: Iron ore reserves estimated over 1,000 sq. km in Sekota to establish sustainable domestic supply.
- **Mining Sector Policy**: Formulation and institutional validation of the comprehensive national Mining Policy.
- **Stakeholder Alignment**: Institutionalization of the National Mining Council, establishing seamless coordination among federal ministries, regional states, and academic institutions.
- **Global Promotion**: Continuous execution of annual International Mining Expos, raising international visibility and investor interest.
- **Environmental Hazard Clearance**: Safe, scientifically compliant disposal of hazardous laboratory chemicals accumulated over decades.
- **Regional Empowerment**: Delivery of technical, analytical, and logistical equipment packages to regional mining administrations.

### 2.6.2 Challenges Encountered

- **Security and Stability Risks**: Localized security challenges in specific mineral-rich extraction localities.
- **Technical Skills Deficits**: Shortage of specialized, highly trained human resources and modern exploration technologies.
- **Production Deficits**: Concessionaires failing to attain contractual output thresholds due to logistical, technical, or financial hurdles.
- **Inter-Regional Disparities**: Lack of uniform organizational structures and administrative standards across regional mining offices.
- **Infrastructure Bottlenecks**: Deficits in road access, national electricity grid connectivity, and industrial water supply in remote deposits.
- **Field Logistics Constraints**: Severe shortage of operational field vehicles required for rigorous on-site compliance audits.
- **Absence of Integrated Digital Tracking**: Incomplete rollout of digital systems for contract administration and real-time monitoring.
- **Knowledge Gaps**: Insufficient nationwide modern airborne geophysical coverage relative to the country's vast mineral endowment.

- **Gemstone Value-Chain Gaps**: Technological limitations in certified laboratory grading, cutting, and international valuation.

### 2.6.3 Solutions Taken to Address Challenges

- Close, continuous coordination mechanisms established between federal security forces, regional governments, and local communities to ensure peaceful mining environments.
- Dedicated inter-agency steering task forces created to provide hands-on troubleshooting and operational facilitation for mega mining enterprises.
- Collaborative initiatives launched with universities, the Information Network Security Administration (INSA), and the Ethiopian Commodity Exchange (ECX) to modernize service delivery, digitalize mineral cadastre licensing, and establish structured commodity trading systems.

# Part Three: Medium-Term (2019-2021 E.C.) Development Plan Implications and Expected Results

This Medium-Term Plan harmonizes sector priorities with the overarching national 10-Year Perspective Development Plan. The plan focuses heavily on maximizing foreign currency generation, replacing imported industrial raw materials and finished goods with domestic resources, fostering extensive employment, and driving sustainable local community development.

## 3.1 Focus Directions for Implementation (2019-2021)

- Strengthening institutional capacity-building and human capital development across the entire mining value chain.
- Expanding the volume, diversity, and quality of export minerals to achieve substantial foreign exchange earnings.
- Accelerating domestic substitution of imported industrial mineral feedstocks and finished goods.
- Enhancing capacity utilization, efficiency, and productivity among mineral processing and manufacturing industries.
- Modernizing public service delivery through end-to-end digital transformation and e-governance systems.
- Extending structured technical, logistical, and formalization support to enhance the productivity of artisanal miners.
- Executing high-impact domestic and international promotional campaigns to attract premier global exploration and mining corporations.
- Strengthening project support and monitoring systems to enforce strict contractual compliance and project performance.
- Maintaining synchronized inter-agency facilitation to ensure mega mining, petroleum, and geothermal projects operate at full capacity.

- Generating high-quality geological, geochemical, and geophysical datasets using cutting-edge technologies to drive Foreign Direct Investment.
- Ensuring mining projects deliver equitable local community benefits, uphold strict occupational safety, and execute mandatory post-mining land rehabilitation.
- Deepening coordinated partnerships with regional administrations, federal entities, and key stakeholders.

## 3.2 Plan Assumptions

- The federal government will sustain and reinforce its strategic priority and political commitment to the mining sector.
- Public service delivery and regulatory workflows will fully transition into modern digital systems.
- Enabling sector policies and revised legal frameworks will be enacted promptly.
- Targeted capacity-building training and performance incentive packages will be actively implemented.
- Pre-production mega enterprises will ramp up smoothly to achieve full commercial production capacity.
- Competitive gold purchasing incentive frameworks will be maintained in responsiveness to prevailing market conditions.
- International market prices for gold, critical minerals, and industrial commodities will maintain a favorable trajectory.
- Critical public infrastructure (roads, electric power, water) will be progressively expanded in mining corridors.
- Localized security constraints in extraction areas will be decisively addressed and resolved.
- Appropriate capital budgets will be timely allocated by the federal treasury to finance priority projects.
- Regional governments and key institutional partners will maintain decisive commitment and support for sector objectives.

## 3.3 Main Activities and Expected Results in the Medium Term

The mining sector plays an irreplaceable role in driving Ethiopia's rapid macroeconomic expansion. Under this Medium-Term Plan, the Ministry of Mines, in concert with stakeholders, is undertaking extensive initiatives to transform the sector into a leading engine of national capital accumulation. These initiatives focus on deploying advanced technologies to intensify geological investigations, executing targeted investment promotion campaigns, optimizing legal and incentive frameworks to attract domestic and foreign direct investment, expanding export volume and variety, institutionalizing value-addition systems for precious stones, substituting imported construction and industrial inputs, building the capacity of artisanal producers, and creating broad-based employment.

## 3.4 Medium-Term (2019-2021) Strategic Goals and Main Activities

### Strategic Goal 1: Ensure the Country's Geological Information Coverage, Quality, and Accessibility

- **Main Activity 1**: Elevate geoscience, geochemical, geophysical, and geological hazard mapping coverage and data quality.
- **Main Activity 2**: Expand exploration and investigations for mineral inputs enhancing agricultural productivity.
- **Main Activity 3**: Broaden exploration and studies for infrastructure, construction, and industrial mineral inputs.
- **Main Activity 4**: Accelerate exploration and feasibility studies of strategic export-oriented minerals.
- **Main Activity 5**: Modernize geoscience laboratory analysis and public information dissemination services.
- *Expected Results*: Completed geoscience mapping sheets; delineated mineral prospective zones; evaluated mineral deposits; certified laboratory sample analyses; and accessible digital geoscience archives.

### Strategic Goal 2: Enhance Private Sector Investment and Mining Industry Production Capacity

- **Main Activity 1**: Expand domestic and foreign private investment in mining and petroleum development.
- **Main Activity 2**: Enhance operational capacity utilization across mineral processing industries.
- **Main Activity 3**: Accelerate overall output across mineral and petroleum industries.
- *Expected Results*: Substantial registered investment capital; issued enterprise licenses; active developers in petroleum and geothermal energy; elevated industrial capacity utilization; and expanded domestic industrial output.

### Strategic Goal 3: Expand Mining Sector Production, Foreign Exchange Receipts, and Job Creation

- **Main Activity 1**: Maximize production and export sales of foreign-exchange-generating minerals.
- **Main Activity 2**: Expand domestic production of industrial minerals and petroleum products.
- **Main Activity 3**: Mobilize domestic tax, royalty, and service revenues from the mining sub-economy.
- **Main Activity 4**: Generate broad-based employment opportunities across industrial and artisanal mining.
- *Expected Results*: Higher mineral export volumes; enhanced artisanal productivity; substituted import volumes; expanded foreign exchange receipts and savings; increased government revenue; and extensive sustainable employment.

### Strategic Goal 4: Strengthen Community Linkages and Environmental Protection in Mining and Petroleum Investment

- **Main Activity 1**: Mobilize and disburse local community development funds from concessionaires.

- *Expected Results*: Mobilized and invested community development funds; fully restored post-mining ecosystems; heightened community ownership; and peaceful operating environments.

### 3.5 Implementation Strategies to Achieve the Goals

- Deliver specialized skills training in-country and abroad across critical disciplines (Mineral Economics, Marketing, Risk Management, Promotion and Negotiation, Licensing Frameworks, Digital Compliance, Project M&E).
- Review developers' work programs and conduct rigorous field inspections to guarantee contractual execution.
- Identify developers' emerging operational challenges and resolve bottlenecks collaboratively with key stakeholders.
- Extend logistical, technological, and formalization support to artisanal and small-scale miners through targeted projects.
- Collaborate with regional administrations to formalize artisanal and small-scale mining cooperatives.
- Establish additional formal gold purchasing centers in regional mining clusters lacking official buying stations.
- Transfer thoroughly explored mineral prospective sites to organized artisanal mining associations.
- Deliver technical training to gemstone producers on advanced extraction, cutting, and polishing techniques.
- Provide dedicated supply chain and market oversight for metallic, industrial, and decorative minerals.
- Delineate strategic mineral corridors across the nation and enforce quality standards from extraction to market.
- Conclude the integration of precious gemstone transactions into the modern trading platform of the Ethiopian Commodity Exchange (ECX).
- Execute joint operations with security agencies and customs authorities to combat illicit mineral trafficking and contraband smuggling.
- Track gold volumes arriving at central purchasing stations and supply real-time intelligence to decision-makers.
- Conduct continuous environmental audits to ensure mining projects benefit local communities and execute required ecological rehabilitation.
- Deploy the *Digital Realtime Monitoring and Evaluation System* to oversee mining, petroleum, and geothermal concessions.
- Establish the *Digital Mineral Marketing Information System* to track dynamic international pricing and market trends.
- Roll out the *Mineral Cadastre Information System* to enhance transparency and efficiency in license administration.

# Part Four: Medium-Term (2019-2021 E.C.) Development

# Plan Goals, Main Activities, and Targets

## Detailed Description of Strategic Goals

**Strategic Goal 1: Ensuring Geological Information Coverage, Quality, and Accessibility**: Focuses on establishing a comprehensive digital geological repository, modernizing exploration via airborne geophysics, partnering with the Ministry of Agriculture on soil acidity mapping and limestone resource clustering, discovering industrial feedstocks to replace imports, prioritizing high-value critical minerals, and establishing online access to national geoscience records for investors.

**Strategic Goal 2: Enhancing Private Investment and Industrial Production Capacity**: Directly tied to national macroeconomic reforms and the revised sector policy. Concentrates on promotional campaigns, digitalizing licensing workflows, streamlining approvals, drafting investment-friendly legal frameworks, accelerating technology transfer, and driving capacity utilization across cement, ceramics, steel, and glass manufacturing plants.

**Strategic Goal 3: Expanding Production, Foreign Exchange Inflows, and Job Creation**: Centers on optimizing commercial output across commissioned mega mines, expanding gold deliveries to the central bank, curbing contraband smuggling, modernizing gemstone marketing via digital portals, enhancing domestic value addition, and creating sustainable employment opportunities across mining corridors.

**Strategic Goal 4: Strengthening Community Linkages and Environmental Sustainability**: Aims at mitigating social and ecological impacts, ensuring host communities participate equitably in developmental gains, enforcing mandatory contributions to community development funds (health posts, water, schools, roads), and executing compulsory post-mining land rehabilitation.

## Medium-Term Key Performance Indicators and Targets Matrix (2019-2021 E.C.)

| No. | Strategic Goals, Main Activities, and Key Performance Indicators | Unit of Measurement | 2018 Base | 2019 | 2020 | 2021 |
|---|---|---|---|---|---|---|
| **1** | **Ensuring the Country's Geological Information Coverage, Quality, and Accessibility** | | | | | |
| ***1.1*** | ***Enhancing Geoscience, Geochemistry, Geophysics, and Geohazards Coverage, Quality, and Accessibility*** | | | | | |
| 1.1.1 | Fundamental geoscience mapping at 1:50,000 scale (Number of sheets) | Number | 4 | 4 | 4 | 4 |
| 1.1.2 | Fundamental geoscience mapping coverage at 1:50,000 scale | Sq. Km | 6,000 | 6,000 | 6,000 | 6,000 |
| 1.1.3 | Geothermal prospective sites identified for direct utilization | Number | 2 | 2 | 2 | 2 |
| 1.1.4 | Delineated geological hazard vulnerable sites | Number | 3 | 3 | 4 | 4 |
| 1.1.5 | Engineering geology and geohazard mapping coverage at 1:50,000 scale | Sq. Km | 6,750 | 6,750 | 9,000 | 9,000 |

| No. | Strategic Goals, Main Activities, and Key Performance Indicators | Unit of Measurement | 2018 Base | 2019 | 2020 | 2021 |
|---|---|---|---|---|---|---|
| *1.2* | ***Exploration and Study of Mineral Inputs Enhancing Agricultural Productivity*** | | | | | |
| 1.2.1 | Identification of mineral reserve sites suitable for fertilizer inputs | Number | 1 | 1 | 1 | 1 |
| 1.2.2 | Exploration coverage for fertilizer mineral inputs | Sq. Km | 750 | 750 | 750 | 750 |
| *1.3* | ***Expanding Exploration and Studies for Infrastructure and Construction Industry*** | | | | | |
| 1.3.1 | Chromium / Cobalt prospective exploration and study | Site Count | 2 | 2 | 2 | 2 |
| 1.3.2 | Nickel prospective exploration and study | Site Count | 2 | 2 | 2 | 2 |
| 1.3.3 | Delineated coal prospective sites studied | Site Count | 2 | 2 | 2 | 2 |
| 1.3.4 | Iron ore prospective exploration and reserve estimation studies | Reserve Studies (No.) | 1 | 3 | 3 | 3 |
| *1.4* | ***Expanding Exploration and Studies of Export-Oriented Minerals*** | | | | | |
| 1.4.2 | Studied gemstone prospective sites | Site Count | 2 | 2 | 2 | 2 |
| 1.4.3 | Studied tantalum and lithium prospective sites | Site Count | 2 | 2 | 2 | 2 |
| 1.4.4 | Studied industrial minerals prospective sites | Site Count | 3 | 3 | 3 | 3 |
| *1.5* | ***Expanding Geological Survey Extension Services and Data Accessibility*** | | | | | |
| 1.5.1 | Geoscience laboratory sample testing and analysis performed | Number | 30,065 | 50,154 | 67,745 | 83,400 |
| 1.5.2 | Core drilling exploration executed at designated prospective sites | Meters | 700 | 700 | 1,000 | 1,200 |
| 1.5.3 | Geoscience information sets compiled, organized, and disseminated | Number | 13,000 | 14,000 | 14,500 | 15,000 |
| 1.5.4 | Studied mineral potential sites officially transferred to regional states | Number | 17 | 17 | 17 | 17 |
| **2** | **Enhancing Private Sector Investment and Mining Industry Production Capacity** | | | | | |
| *2.1* | ***Expanding Domestic and Foreign Investment in Mining and Petroleum*** | | | | | |
| 2.1.1 | Capital registered from newly granted mining licenses | Million Birr | 653.472 | 6,500 | 7,000 | 7,500 |

| No. | Strategic Goals, Main Activities, and Key Performance Indicators | Unit of Measurement | 2018 Base | 2019 | 2020 | 2021 |
|---|---|---|---|---|---|---|
| 2.1.2 | Enterprise mining licenses issued by regional administrations | Number | 9,000 | 9,450 | 9,920 | 10,416 |
| 2.1.3 | Geothermal exploration and development licenses issued | Number | 2 | 2 | 3 | 4 |
| 2.1.4 | Capital registered from geothermal investment permits | Million USD | 3.5 | 5.0 | 7.0 | 10.0 |
| 2.1.5 | Geothermal well-drilling exploration permits issued | Number | 2 | 3 | 4 | 4 |
| 2.1.6 | Petroleum development licenses issued | Number | 1 | 1 | 1 | 1 |
| ***2.2*** | ***Problem-Solving Studies Enhancing Industrial Mining Capacity*** | | | | | |
| 2.2.1 | Cement industrial capacity utilization rate | Percentage (%) | 67% | 68% | 70% | 72% |
| 2.2.2 | Ceramic industrial capacity utilization rate | Percentage (%) | 65% | 67% | 69% | 71% |
| 2.2.3 | Steel industrial capacity utilization rate | Percentage (%) | 15.44% | 15.5% | 16.7% | 18% |
| 2.2.4 | Marble and granite processing capacity utilization rate | Percentage (%) | 50% | 55% | 60% | 65% |
| 2.2.4b | Sheet glass manufacturing capacity utilization rate | Percentage (%) | 60% | 60.5% | 62% | 64% |
| 2.2.5 | Chemical manufacturing capacity utilization rate | Percentage (%) | 55% | 58.5% | 61.5% | 65% |
| 2.2.6 | Problem-solving R&D studies conducted for mineral input consumers | Number | 8 | 8 | 9 | 9 |
| ***2.3*** | ***Expanding Mining Industrial Production*** | | | | | |
| 2.3.1 | Industrial cement production volume | Million Tons | 10.60 | 11.66 | 12.80 | 14.08 |
| 2.3.2 | Industrial steel production volume | Million Tons | 0.504 | 0.554 | 0.609 | 0.669 |
| 2.3.3 | Industrial ceramic tile production volume | Million Sq. M | 13.97 | 22.52 | 28.75 | 38.51 |
| 2.3.4 | Sheet glass production volume | Sq. Meters | 1,251,000 | 1,376,000 | 1,514,000 | 1,665,000 |
| 2.3.5 | Industrial marble production volume | Thousand Sq. M | 87.70 | 96.40 | 106.00 | 116.60 |
| 2.3.6 | Industrial chemical products volume | Metric Tons | 5,010 | 5,511 | 6,062.1 | 6,668.3 |
| 2.3.7 | Weed and pest control chemical manufacturing volume | Liters | 142,942.5 | 157,236.75 | 172,960.42 | 190,256.46 |

| No. | Strategic Goals, Main Activities, and Key Performance Indicators | Unit of Measurement | 2018 Base | 2019 | 2020 | 2021 |
|---|---|---|---|---|---|---|
| 2.3.8 | Agricultural lime production volume for soil treatment | Million Quintals | 0.702 | 0.773 | 0.850 | 0.935 |
| 2.3.9 | Industrial salt production volume | Metric Tons | 405,261.43 | 425,524.50 | 446,800.70 | 469,140.80 |
| 2.3.10 | Domestic coal production volume | Million Tons | 1.97 | 2.16 | 2.37 | 2.60 |
| 2.3.11 | Granite production volume | Million Tons | 189.60 | 210.70 | 231.70 | 254.80 |
| 2.3.12 | Terrazzo tile production volume | Thousand Sq. M | 108.40 | 119.20 | 131.12 | 144.20 |
| 2.3.13 | Construction mineral raw materials supply volume | Million Cu. M | 36.60 | 38.43 | 40.35 | 42.37 |
| 2.3.14 | Industrial mineral inputs supply volume | Million Tons | 9.29 | 9.76 | 10.24 | 10.76 |
| 2.3.15 | Natural gas production volume | Thousand Metric Tons | 50 | 120 | 120 | 360 |
| 2.3.16 | Thermal power generated from natural gas | Megawatts (MW) | - | 1,000 | - | - |
| 2.3.17 | Electric power generated from geothermal resources | Megawatts (MW) | - | 5 | - | 10 |
| **3** | **Expanding Mining Sector Output, Foreign Exchange, and Employment** | | | | | |
| ***3.1*** | ***Increasing Export Mineral Production*** | | | | | |
| 3.1.1 | Gold export production volume | Metric Tons | 43.23 | 47.55 | 52.30 | 57.53 |
| 3.1.2 | Tantalum export production volume | Metric Tons | - | - | - | - |
| 3.1.3 | Gemstone export production volume | Metric Tons | 18.00 | 20.10 | 22.11 | 24.32 |
| 3.1.4 | Opal export production volume | Metric Tons | 8.886 | 9.774 | 10.749 | 11.823 |
| 3.1.5 | Lithium export production volume | Metric Tons | - | - | - | - |
| 3.1.6 | Export-quality marble production volume | Cubic Meters | 472.69 | 519.96 | 571.92 | 629.12 |
| 3.1.7 | Sodium bromide mineral production volume | Metric Tons | 5,200 | 5,720 | 6,292 | 6,921 |
| 3.1.8 | Industrial minerals export shipment volume | Metric Tons | 27,573.75 | 30,331.12 | 33,364.23 | 36,700.60 |
| 3.1.9 | Foreign exchange revenue generated from mineral exports | Billion USD | 5.20 | 5.72 | 6.20 | 6.80 |
| ***3.3*** | ***Mobilizing Domestic Sector Revenue*** | | | | | |
| 3.3.1 | Royalty revenue collected from mineral extraction | Million Birr | 1,052.00 | 1,100.00 | 1,200.00 | 1,300.00 |

| No. | Strategic Goals, Main Activities, and Key Performance Indicators | Unit of Measurement | 2018 Base | 2019 | 2020 | 2021 |
|---|---|---|---|---|---|---|
| 3.3.2 | Revenue from licensing, registration, renewal, and competency fees | Million Birr | 2.01 | 2.57 | 3.00 | 3.10 |
| 3.3.3 | Geothermal service revenue collected | Million Birr | 0.015 | 0.025 | 0.030 | 0.035 |
| 3.3.4 | Miscellaneous petroleum regulatory fees collected | Million USD | 0.35 | 0.35 | 0.45 | 0.45 |
| 3.3.5 | Regional mining operations revenue collected | Billion Birr | 3.20 | 3.52 | 3.87 | 4.26 |
| ***3.4*** | ***Job Creation in the Mining Sector*** | | | | | |
| 3.4.1 | Formal employment created within mining companies and industries | Number | 2,000 | 2,130 | 2,605 | 3,100 |
| 3.4.2 | Employment generated in regional artisanal and small-scale mining | Number | 171,590 | 180,170 | 190,000 | 199,500 |
| 3.4.3 | Appropriate technologies deployed for women artisanal miners | Number | - | 1 | 1 | 2 |
| 3.4.4 | Women artisanal miners provided with occupational health & safety gear | Number | - | 200 | 300 | 500 |
| **4** | **Strengthening Community Linkages in Mining and Petroleum Investment** | | | | | |
| ***4.1*** | ***Mobilizing Community Development Funds from Developers*** | | | | | |
| 4.1.1 | Community development funds mobilized from concessionaires | Million Birr | 43.00 | 55.00 | 65.00 | 70.00 |
| 4.1.3 | Community development funds disbursed to regional administrations | Million Birr | 42.00 | 55.00 | 65.00 | 70.00 |
| 4.1.4 | Post-mining land area rehabilitated and restored | Hectares | 45 | 50 | 55 | 60 |

## 4.1.1 Medium Term Development and Investment in Million from 2019-2021 E.C.

> *Capital Expenditure Allocation Matrix (Values in Millions of Ethiopian Birr, funded via the Federal Treasury).*

| Program | Sub-Program / Institutional Program | 2019 | 2020 | 2021 | Total 2019-2021 | Impl. Body | Funding Source | Region |
|---|---|---|---|---|---|---|---|---|
| **Administration and Management Program** | Providing Support and Service | 440.81 | 499.04 | 546.00 | **1,485.85** | All | Federal Treasury | Addis Ababa |
| **Geoscience and Mineral Exploration** | Increasing the coverage, quality, and accessibility of geoscience data | 368.43 | 267.98 | 220.91 | **857.32** | GIE | Federal Treasury | Nationwide |
| | Increasing exploration and studies for agricultural enhancement; expanding exploration for infrastructure, construction, industrial inputs, and export markets | 14,531.92 | 19,912.18 | 6,326.22 | **40,770.32** | GIE | Federal Treasury | Nationwide |
| | Geo-Science Services | 1,549.80 | 1,040.81 | 791.65 | **3,382.27** | GIE | Federal Treasury | Addis Ababa |
| **Mining Industry Growth and Competitiveness** | Enhancing Mining Sector Investment & Strengthening Local Community Engagement in Mineral Sector Development | 52.38 | 457.61 | 513.37 | **1,023.36** | MoM | Federal Treasury | Nationwide |
| **Mining Industry Research and Development** | Expanding the mining industry product and improving productivity | 433.62 | 1,208.27 | 1,559.95 | **3,201.84** | MIDI | Federal Treasury | Nationwide |

| Program | Sub-Program / Institutional Program | 2019 | 2020 | 2021 | Total 2019-2021 | Impl. Body | Funding Source | Region |
|---|---|---|---|---|---|---|---|---|
| **Promoting Mineral Export and Import Substitution** | Enhancing the export of mineral products; enhancing mineral industry and petroleum production; increasing domestic revenue; expanding employment opportunities | 620.50 | 207.83 | 184.81 | **1,013.14** | MoM | Federal Treasury | Nationwide |
| **Total Medium-Term Capital Investment** | | **17,997.46** | **23,385.72** | **9,942.91** | **51,734.10** | **Federal Treasury (All Regions)** | | |

## 4.1 Sector Strengths, Weaknesses, Opportunities, and Threats (SWOT Analysis)

### 4.1.1 Strengths (Strength)

- Active, ongoing programs to enhance sector human capital and technical specialization.
- Established technology-enabled licensing administration systems.
- Implementation of the Electronic Single Window trading and facilitation system.
- Strategic capital projects operationalized to enhance institutional execution capacity.
- Development of customized software applications and modern IT infrastructure.
- Favorable, productive institutional working environment.
- Successful commercial commissioning and entry of mega mining companies.

### 4.1.2 Weaknesses (Weakness)

- Gaps in institutional implementation, enforcement, and monitoring capacity.
- Incomplete operationalization of a fully integrated digital contract administration and compliance tracking system.

### 4.1.3 Opportunities (Opportunities)

- Vast, largely untapped mineral, petroleum, and geothermal natural endowments across Ethiopia.
- High-level political commitment and strategic priority accorded to the sector by the federal government.
- Nationwide digital transformation of public institutions and services.
- Comprehensive national macroeconomic reforms incentivizing formal foreign exchange earnings.
- Rising international market demand and favorable pricing for gold and critical minerals.

- Growing interest and entry of highly capitalized domestic and global mining corporations.
- Establishment of the high-level National Mining Steering Committee.
- Implementation of broad public service and administrative civil service reforms.

### 4.1.4 Threats (Threat)

- Localized security and stability risks across specific mineral extraction corridors.
- Shortage of certified, experienced technical professionals across specialized geoscience disciplines.
- High financial expenditure requirements for modern exploration (e.g., airborne geophysics) relative to available budgets.
- Varying organizational capacity and lack of structural uniformity across regional mining bureaus.
- Budgetary constraints impacting the timely execution of public capital projects.
- Infrastructure deficits (access roads, power grid, water supplies) in remote mineral deposit areas.
- Fuel shortages impacting exploration rigs and mining heavy machinery.

### 4.1.5 Strategies to Mitigate Threats

- Strengthen and sustain synchronized joint security task forces between federal and regional security agencies in extraction corridors.
- Facilitate comprehensive short- and long-term domestic and overseas training programs for sector professionals.
- Deepen collaborative coordination with regional administrations, federal stakeholders, and developer associations.

# Part Five: Monitoring and Evaluation System to be Implemented during the Investment Plan Period (2019-2021 E.C.)

A structured Monitoring and Evaluation (M&E) system is the primary management instrument for elevating program performance and guaranteeing that expected developmental outcomes are realized. The Ministry of Mines has prepared and operationalized a comprehensive Planning, Monitoring, and Evaluation Manual. Through timely monitoring and institutional support, the system holds stakeholders accountable to their mandates, gathers operational field intelligence to calibrate subsequent plans, facilitates evidence-based decision-making, and guarantees that the sector fulfills its government- and citizen-mandated mission.

Building upon lessons learned from past monitoring exercises—specifically addressing gaps such as fragmented inter-agency coordination, lack of modern digital M&E tools, over-emphasis on procedural activities rather than verifiable impacts, and the absence of institutional accountability frameworks—a rigorous, results-based M&E framework will be deployed across all institutional levels based on tangible performance indicators:

- **Paperless Digital Data Collection**: Deploy an integrated digital database system to gather, process, and analyze real-time operational data without paper touchpoints, providing real-time data access to

executive decision-makers.

- **Accountability and Feedback Mechanism**: Institute an actionable performance feedback system that rewards exemplary achievements while enforcing administrative accountability where compliance deficits occur.
- **Dynamic Annual Plan Alignment**: Calibrate annual operational plans against this Medium-Term Investment Plan, conducting periodic reviews to incorporate emerging market realities.
- **Internal and External Consensus Building**: Conduct regular consultative forums with Ministry staff and key institutional stakeholders to maintain shared operational alignment.
- **Rigorous Contractual Compliance Audits**: Conduct multi-tiered reviews of progress reports and on-site field inspections to verify that licensed concessionaires honor their contractual work programs and expenditure commitments.
- **Joint High-Level Field Missions**: Organize regular field inspection missions uniting Ministry senior executives and regional leadership to directly inspect operations and resolve developer challenges on the ground.
- **Synchronized Anti-Contraband Operations**: Execute coordinated intelligence-led enforcement operations with federal security agencies, customs authorities, and regional administrations to eliminate illegal mineral trading and contraband smuggling.
- **Community Benefit and Environmental Audits**: Rigorously track developers' compliance with local community development fund contributions and enforce mandatory post-mining ecosystem rehabilitation (Rehabilitation).
- **Periodic Multi-Stakeholder Evaluation Platforms**:
  - Semi-annual joint on-site field monitoring and performance evaluation conducted with the Parliamentary Standing Committee on Industry and Mining Affairs.
  - Quarterly consultative performance review forums uniting developers, government entities, and relevant stakeholders, supplemented by regular operational review sessions.

# Part Six: Conclusion

The mining sector holds indispensable significance for Ethiopia's macroeconomic expansion and structural transformation. Comprehensive reform initiatives are underway to ensure the nation's natural endowments are utilized at an elevated scale. Factoring in these achievements, this plan has been formulated to accelerate Ethiopia's prosperity journey over the next three years (2019-2021 E.C.) and enable the mining sector to fulfill its transformative role.

The plan centers on deploying cutting-edge technologies to explore mineral reserves, attracting high-caliber investment, elevating industrial productivity, modernizing public services through digital platforms, and establishing benchmark operational achievements. To realize these strategic objectives, concerted efforts, synchronized multi-stakeholder partnerships across federal and regional institutions, and rigorous monitoring and support systems will be mobilized to drive superior sector performance.